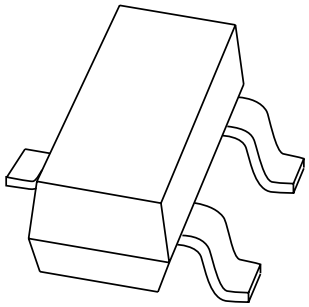
# DATA SHEET



# PMBT2907; PMBT2907A
PNP switching transistors

Product specification
Supersedes data of 1997 Sep 04

1999 Apr 27

Philips
Semiconductors

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# PNP switching transistors PMBT2907; PMBT2907A

## FEATURES

- High current (max. 600 mA)
- Low voltage (max. 60 V).

## APPLICATIONS

- Switching and linear amplification.

## DESCRIPTION

PNP switching transistor in a SOT23 plastic package.
NPN complements: PMBT2222 and PMBT2222A.

## MARKING

| TYPE NUMBER | MARKING CODE(1) |
|---|---|
| PMBT2907 | ∗2B |
| PMBT2907A | ∗2F |

**Note**

1. ∗ = p : Made in Hong Kong.
   ∗ = t : Made in Malaysia.

## PINNING

| PIN | DESCRIPTION |
|---|---|
| 1 | base |
| 2 | emitter |
| 3 | collector |

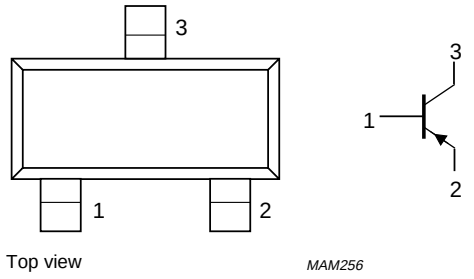



Fig.1 Simplified outline (SOT23) and symbol.

## LIMITING VALUES

In accordance with the Absolute Maximum Rating System (IEC 134).

| SYMBOL | PARAMETER | CONDITIONS | MIN. | MAX. | UNIT |
|---|---|---|---|---|---|
| $V_{CBO}$ | collector-base voltage | open emitter | – | –60 | V |
| $V_{CEO}$ | collector-emitter voltage | open base | | | |
| | PMBT2907 | | – | –40 | V |
| | PMBT2907A | | – | –60 | V |
| $V_{EBO}$ | emitter-base voltage | open collector | – | –5 | V |
| $I_C$ | collector current (DC) | | – | –600 | mA |
| $I_{CM}$ | peak collector current | | – | –800 | mA |
| $I_{BM}$ | peak base current | | – | –200 | mA |
| $P_{tot}$ | total power dissipation | $T_{amb} \leq 25$ °C | – | 250 | mW |
| $T_{stg}$ | storage temperature | | –65 | +150 | °C |
| $T_j$ | junction temperature | | – | 150 | °C |
| $T_{amb}$ | operating ambient temperature | | –65 | +150 | °C |

## THERMAL CHARACTERISTICS

| SYMBOL | PARAMETER | CONDITIONS | VALUE | UNIT |
|---|---|---|---|---|
| $R_{th\ j\text{-}a}$ | thermal resistance from junction to ambient | note 1 | 500 | K/W |

**Note**

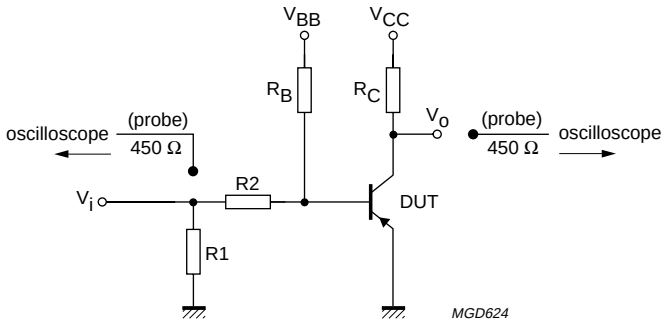
1. Transistor mounted on an FR4 printed-circuit board.

## CHARACTERISTICS

$T_j$ = 25 °C unless otherwise specified.

| SYMBOL | PARAMETER | CONDITIONS | MIN. | MAX. | UNIT |
|---|---|---|---|---|---|
| $I_{CBO}$ | collector cut-off current | $I_E$ = 0; $V_{CB}$ = –50 V | | | |
| | PMBT2907 | | – | –20 | nA |
| | PMBT2907A | | – | –10 | nA |
| | collector cut-off current | $I_E$ = 0; $V_{CB}$ = –50 V; $T_j$ = 125 °C | | | |
| | PMBT2907 | | – | –20 | μA |
| | PMBT2907A | | – | –10 | μA |
| $I_{EBO}$ | emitter cut-off current | $I_C$ = 0; $V_{EB}$ = –5 V | – | –50 | nA |
| $h_{FE}$ | DC current gain | $I_C$ = –0.1 mA; $V_{CE}$ = –10 V | | | |
| | PMBT2907 | | 35 | – | |
| | PMBT2907A | | 75 | – | |
| | DC current gain | $I_C$ = –1 mA; $V_{CE}$ = –10 V | | | |
| | PMBT2907 | | 50 | – | |
| | PMBT2907A | | 100 | – | |
| | DC current gain | $I_C$ = –10 mA; $V_{CE}$ = –10 V | | | |
| | PMBT2907 | | 75 | – | |
| | PMBT2907A | | 100 | – | |
| | DC current gain | $I_C$ = –150 mA; $V_{CE}$ = –10 V | 100 | 300 | |
| | DC current gain | $I_C$ = –500 mA; $V_{CE}$ = –10 V | | | |
| | PMBT2907 | | 30 | – | |
| | PMBT2907A | | 50 | – | |
| $V_{CEsat}$ | collector-emitter saturation voltage | $I_C$ = –150 mA; $I_B$ = –15 mA | – | –400 | mV |
| | | $I_C$ = –500 mA; $I_B$ = –50 mA | – | –1.6 | V |
| $V_{BEsat}$ | base-emitter saturation voltage | $I_C$ = –150 mA; $I_B$ = –15 mA | – | –1.3 | V |
| | | $I_C$ = –500 mA; $I_B$ = –50 mA | – | –2.6 | V |
| $C_c$ | collector capacitance | $I_E = i_e$ = 0; $V_{CB}$ = –10 V; f = 1 MHz | – | 8 | pF |
| $C_e$ | emitter capacitance | $I_C = i_c$ = 0; $V_{EB}$ = –2 V; f = 1 MHz | – | 30 | pF |
| $f_T$ | transition frequency | $I_C$ = –50 mA; $V_{CE}$ = –20 V; f = 100 MHz | 200 | – | MHz |

| SYMBOL | PARAMETER | CONDITIONS | MIN. | MAX. | UNIT |
|---|---|---|---|---|---|
| **Switching times (between 10% and 90% levels);** (see Fig.2) | | | | | |
| $t_{on}$ | turn-on time | $I_{Con}$ = –150 mA; $I_{Bon}$ = –15 mA; $I_{Boff}$ = 15 mA | – | 40 | ns |
| $t_d$ | delay time | | – | 12 | ns |
| $t_r$ | rise time | | – | 30 | ns |
| $t_{off}$ | turn-off time | | – | 365 | ns |
| $t_s$ | storage time | | – | 300 | ns |
| $t_f$ | fall time | | – | 65 | ns |

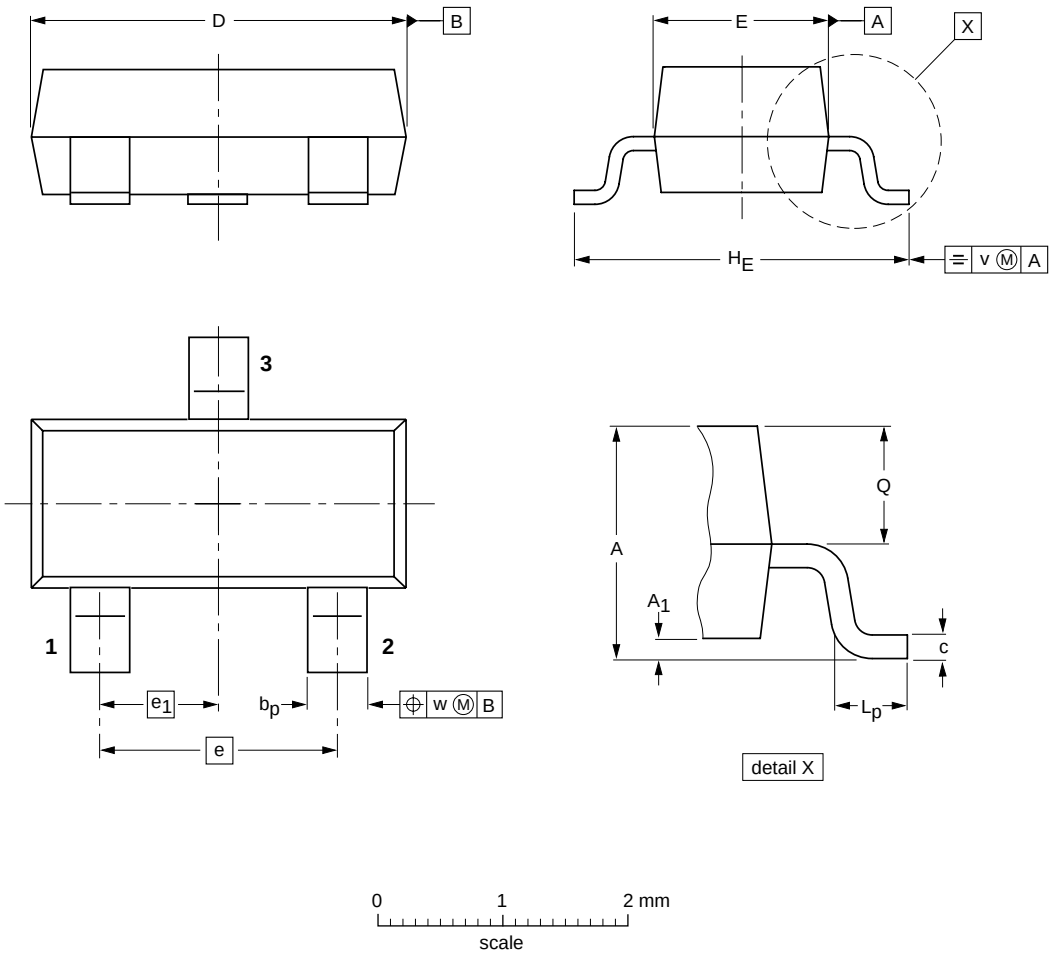



$V_i$ = –9.5 V; T = 500 μs; $t_p$ = 10 μs; $t_r = t_f \leq 3$ ns.
R1 = 68 Ω; R2 = 325 Ω; $R_B$ = 325 Ω; $R_C$ = 160 Ω.
$V_{BB}$ = 3.5 V; $V_{CC}$ = –29.5 V.
Oscilloscope: input impedance $Z_i$ = 50 Ω.

Fig.2 Test circuit for switching times.

## PACKAGE OUTLINE

**Plastic surface mounted package; 3 leads** **SOT23**




**DIMENSIONS (mm are the original dimensions)**

| UNIT | A | $A_1$ max. | $b_p$ | c | D | E | e | $e_1$ | $H_E$ | $L_p$ | Q | v | w |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| mm | 1.1<br>0.9 | 0.1 | 0.48<br>0.38 | 0.15<br>0.09 | 3.0<br>2.8 | 1.4<br>1.2 | 1.9 | 0.95 | 2.5<br>2.1 | 0.45<br>0.15 | 0.55<br>0.45 | 0.2 | 0.1 |

| OUTLINE VERSION | REFERENCES | | | | EUROPEAN PROJECTION | ISSUE DATE |
|---|---|---|---|---|---|---|
| | IEC | JEDEC | EIAJ | | | |
| SOT23 | | | | | | 97-02-28 |

## DEFINITIONS

| **Data sheet status** | |
|---|---|
| Objective specification | This data sheet contains target or goal specifications for product development. |
| Preliminary specification | This data sheet contains preliminary data; supplementary data may be published later. |
| Product specification | This data sheet contains final product specifications. |
| **Limiting values** | |
| Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability. | |
| **Application information** | |
| Where application information is given, it is advisory and does not form part of the specification. | |

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